NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Introduction to Internet of Things

Assignment-Week 3

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 15 **Total marks: 15 X 1= 15**

QUESTION 1:

State True or False.

Statement: "Wireless HART is the latest release of Highway Addressable Remote Transducer protocol."

a. True
b. False

Correct Answer: a. True

Detailed Solution: WirelessHART is the latest release of Highway Addressable Remote Transducer protocol.

See lecture 11 (Connectivity Technologies-III) @ 01:42

QUESTION 2:

State True or False.

Statement: "Wired HART does not have a network layer."

a. True
b. False

Correct Answer: a. True

Detailed Solution: Wired HART does not have a network layer.

See lecture 11 (Connectivity Technologies-III) @ 04:25

QUESTION 3:

Wireless HART physical layer is derived from _____ protocol?

a. 802.15.4
b. 802.15.2
c. 802.15.3
d. 802.15.6

Correct Answer: a. 802.15.4

Detailed Solution: WirelessHART physical layer is derived from 802.15.4 protocol.

See lecture 11 (Connectivity Technologies-III) @ 06:00

QUESTION 4:

Wireless HART operates only in _____ GHz ISM band.

a. 3.7
b. 4.8
c. 4.8
d. None of these

Correct Answer: d. None of these

Detailed Solution: HART operates only in the 2.4 GHz ISM band.

See lecture 11 (Connectivity Technologies-III) @ 06:00

QUESTION 5:

Wireless HART employs and exploits _____ channels of the ISM band to increase reliability.

a. 13
b. 4
c. 2
d. None of these

Correct Answer: d. None of these

Detailed Solution: WirelessHART employs and exploits 15 channels of the ISM band to increase reliability.

See lecture 11 (Connectivity Technologies-III) @ 06:00

QUESTION 6:

State True or False.

Statement: "Channel blacklisting identifies channels consistently affected by interference and adds them from use again."

a. True
b. False

Correct Answer: b. False

Detailed Solution: Channel blacklisting identifies channels consistently affected by interference and removes them from use.

See lecture 11 (Connectivity Technologies-III) @ 07:05

QUESTION 7:

The Wireless HART incorporates _____ and _____ to increase the reliability and security of the network.

a. Channel hopping and Channel Blacklisting
b. Channel Aggregating and Frequency Aggregating
c. Both (a) and (b)
d. Neither (a) nor (b)

Correct Answer: a. Channel hopping and Channel Blacklisting

Detailed Solution: The WirelessHART incorporates Channel hopping and channel blacklisting to increase the reliability and security of the network.

See lecture 11 (Connectivity Technologies-III) @ 07:30

QUESTION 8:

State whether the following statement is True or False

Statement: "Wireless HART protocol relies on Mesh Networking for its communication, and each device is primed to forward packets from every other device.".

a. True
b. False

Correct Answer: a. True

Detailed Solution: WirelessHART protocol relies on Mesh Networking for its communication, and each device is primed to forward packets from every other device.

See lecture 11 (Connectivity Technologies-III) @ 09:00

QUESTION 9:

State True or False.

Statement: "ZigBee protocol does not feature channel hopping at all."

a. True
b. False

Correct Answer: a. True

Detailed Solution: ZigBee protocol does not feature channel hopping at all.

See lecture 11 (Connectivity Technologies-III) @ 14:55

QUESTION 10:

NFC is designed for use by devices within _____ to each other.

a. Close proximity
b. Anywhere on the globe
c. A small building
d. None of these

Correct Answer: a. Close proximity

Detailed Solution: NFC is designed for use by devices within close proximity to each other.

See lecture 11 (Connectivity Technologies-III) @ 17:47

QUESTION 11:

NFC are of _____ types?

a. 2
b. 3
c. 1
d. None of these

Correct Answer: a. 2

Detailed Solution: NFC are of 2 types.

See lecture 11 (Connectivity Technologies-III) @ 18:47

QUESTION 12:

State True or False.

Statement: "Smartphones are an example of active NFC devices."

a. True
b. False

Correct Answer: a. True

Detailed Solution: Smartphones are an example of active NFC devices.

See lecture 11 (Connectivity Technologies-III) @ 18:47

QUESTION 13:

Bluetooth technology is based on _____.

a. HART
b. ZigBee
c. Ad-hoc piconets
d. None of these

Correct Answer: c. Ad-hoc piconets

Detailed Solution: Bluetooth technology is based on Ad-hoc piconets.

See lecture 12 (Connectivity Technologies-IV) @ 02:30

QUESTION 14:

State whether the following statement is true or false.

Statement: The Link Manager Protocol in Bluetooth manages the establishment, authentication, link configuration.

a. True
b. False

Correct Answer: a. True

Detailed Solution: The Link Manager Protocol in Bluetooth manages the establishment, authentication, link configuration.

See Page number – 157, Chapter - 7, Book - Introduction to IoT, Authors – Sudip Misra, Anandarup Mukherjee, and Arijit Roy, Publisher – Cambridge University Press, Edition – 1 (2021)

QUESTION 15:

Zigbee commonly uses _____data rate.

a. 250 Mbps
b. 260 kbps
c. 260 Mbps
d. None of these

Correct Answer: d. None of these

Detailed Solution: Zigbee commonly uses a 250 kbps data rate.

See Page number – 131, Chapter - 7, Book - Introduction to IoT, Authors – Sudip Misra, Anandarup Mukherjee, and Arijit Roy, Publisher – Cambridge University Press, Edition – 1 (2021)

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